Debartha Paul

PhD Student in Statistics
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Education

Doctor of Philosophy, Statistics *Aug'24 – Jul'29 (expected)*
Iowa State University, Ames, IA
Advisor: Dr. Kyle Schindl

Master of Science, Statistics *Sep'22 – Jul'24*
Presidency University, Kolkata, India

Bachelor of Science, Statistics (Hons.) *Aug'19 – Jun'22*
Presidency University, Kolkata, India

Research

My research focuses primarily on problems in Causal Inference, Optimal Transport and Semiparametric Efficiency.

Manuscripts under Preparation:

- On the Geometry of Distributional Treatment Effect Heterogeneity *with Kyle Schindl*

Presentations

- Student Poster Competition (selected), IISA 2026, Varanasi, India (December 2026)

Teaching Experience

Iowa State University *Aug'24 – May'25 & Aug'26 – Present*
Graduate Teaching Assistant

- Grader, Statistical Methods III (STAT 5503, *formerly STAT 5200*), Fall'26.
- Lab instructor and grader, Intermediate Statistical Concepts and Methods (STAT 3201, *formerly STAT 3010*), Fall'26.
- Grader, Time Series Data Analysis (STAT 4720/5720), Spring'25.
- Lab instructor and grader, Intermediate Statistical Concepts and Methods (STAT 3010), Fall'24.

Professional Experience

Corteva Agriscience *Aug'25 – Aug'26*
Student Consultant, Graduate Research Assistant (Remote)

- Identified an optimization bottleneck in an enterprise adaptive-experimentation pipeline where the algorithm reliably converged to a constrained local maximum rather than the global optimum.
- Resolved it by augmenting the search space with a linearly dependent dimension, adding degrees of freedom that let the optimizer escape the constrained maximum and close roughly 90% of the remaining gap to the best known solution.
- Built a Python framework to evaluate **Bayesian Optimization** approaches against baselines on a simulated high-dimensional response surface.

- Assessed convergence and stability through parallelized multi-seed simulations, using bootstrap confidence bands to quantify the performance gain over baseline.

MSA Labs *Apr'24 – Jun'24*

Data Science Intern (Remote)

- Developed instructional materials teaching pharmacometrics practitioners to use Julia for statistical analysis.
- Created worked examples and tutorials covering statistical-computing workflows in Julia for a non-programming audience.

Academic Projects

Probability-based Response-adaptive Allocation Design in Crossover Trials *Jun'24*

Masters dissertation at Presidency University, Kolkata. Supervised by ***Dr. Saurav De***

- Proposed and developed an allocation design for crossover clinical trials that are both response-adaptive and probability-based.
- Used a simulation study to verify our allocation and also find equivalence tests for the treatments.

GitHub

Some Problems on Dice, Their Mathematical Treatments and Simulations *Nov'22*

Joint work with ***Ahel Kundu***

- Derived closed-form solutions to ten classical dice problems using recursive expectation equations, inclusion–exclusion, and absorbing Markov chains.
- Computed expected hitting times via the fundamental matrix $(I - Q)^{-1}$ and solved an optimal-stopping problem for maximizing expected score.
- Verified each analytical result through Monte Carlo simulation in R, with reproducible seeds and convergence plots.

GitHub ·R-bloggers

Relevant Coursework

Theory: Measure Theory, Advanced Probability Theory, Advanced Statistical Inference
Methods: Linear Models and Design of Experiments, Advanced Statistical Modelling
Electives: Time Series Analysis, Advanced Bayesian Methods, Nonparametric Methods, Stochastic Processes in Modern Machine Learning

Open-Source Contributions

R Project Sprint, 2023 ·Link *Aug'23*

Contributor, base R (Remote)

- Selected to contribute to base R alongside R Core members.
- Contributed documentation fixes and translations for the language.

`SimpleLinReg.jl` ·GitHub *2023*

Built a Julia package end-to-end, including package structure, CI, test coverage, and documentation, as an exercise in the full development workflow.

Skills

Languages: R, Python, Julia, C, C++
Tools & Workflow: Git, LaTeX, Quarto, Excel

Methods: Causal inference, Bayesian methods, Bayesian optimization, Time series, Nonparametric inference, A/B testing, Experimental design
Spoken Languages: English, Bengali, Hindi

Awards & Honors

International Youth Math Challenge *2022 & 2023*
Finalist, Silver Honour in both editions.